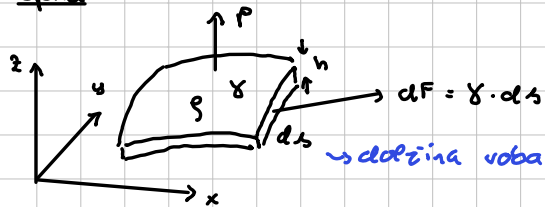


Opna



Osnovni gibalni:

$$\Delta G_z = \int_{t_1}^{t_2} \iint_D z_{tt}(x, y, t) \cdot \frac{dx dy}{ds} \cdot h \rho dt = \int_{t_1}^{t_2} F_z dt + \int_{t_1}^{t_2} \iint_D \rho \frac{dx dy}{ds} \cdot dt$$

↓

spem. giba. količina

↪

↑

→ direk. sila

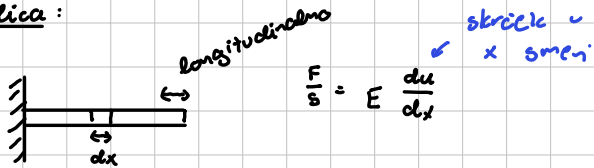
$$F_z = \int_{\partial D} \dots ds$$

↓

Stokes: $\int_D \dots dx dy$

$$z_{tt} = c^2 \nabla^2 z + \frac{\rho}{\gamma h}, \quad c^2 = \frac{\gamma}{\rho h}$$

Palica:



... $\Rightarrow c^2 u_{xx} = u_{tt}, \quad c^2 = \frac{E}{\rho}$

Zvuk: hidrodinamika

Navier - Stokes :

$$\rho \left(\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla) \vec{v} \right) = \underbrace{-\nabla p + \eta \nabla^2 \vec{v} + \vec{g}}_{\text{sile}}$$

Kalor $\downarrow$
 viskoznost $\downarrow$
 zunanje sile (npr. brez) $\swarrow$

$$\frac{\partial \rho}{\partial t} + \nabla (\rho \vec{v}) = 0 \quad \text{kontinuitetna en.}$$

$$\rho = \rho(h) \rightarrow \text{gostota } \rho_{\text{un.}} \text{ h}$$

"majhne" hitrosti, nelinearni $\rightarrow 0$

$$\frac{v^2/\lambda}{v\omega} = \frac{v}{c} \ll 1$$

karakteristične dolžine